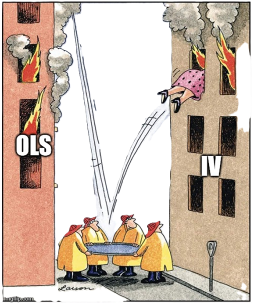
Instrumental Variable Approaches

F. Candau (UPPA-TREE)



The endogeneity problem

We want the **causal** effect β of a treatment x on an outcome y:

$$y_i = \alpha + \beta\, x_i + \varepsilon_i.$$

OLS recovers β only if x_i is uncorrelated with everything else in ε_i. It **fails** whenever $\mathrm{Cov}(x_i, \varepsilon_i) \neq 0$:

- **Omitted variables —** a confounder drives both x and y.
- **Reverse causality / simultaneity —** y also affects x (supply *and* demand).
- **Measurement error** in x (attenuation).

In that case

$$\operatorname{plim} \hat{\beta}_{OLS} = \beta + \frac{\mathrm{Cov}(x_i, \varepsilon_i)}{\mathrm{Var}(x_i)} \neq \beta.$$

Roadmap

Where we are going today:

1. **Instrumental variables** — the two conditions, the estimator, and diagnostics.
2. **Generalized Method of Moments (GMM)** — the framework behind 2SLS and behind the shares view of shift-share.
3. **The shift-share (Bartik) instrument** — what it is and how it is built.
4. **Identification "from the shares"** — Goldsmith-Pinkham et al. (2020).
5. **Identification "from the shifts"** — Borusyak et al. (2022).
6. **Inference** — Adão et al. (2019).
7. **A practical guide** — which assumption fits your design?
8. **Applied session** — estimating a Bartik design in Python.

Two readings underpin the whole lecture: Goldsmith-Pinkham et al. (2020) on shares, Borusyak et al. (2022) on shifts.

Section 1

Instrumental Variables

The IV solution: two conditions

An **instrument** z is a variable that moves x but is otherwise unrelated to the outcome. It must satisfy two conditions:

Relevance (testable)

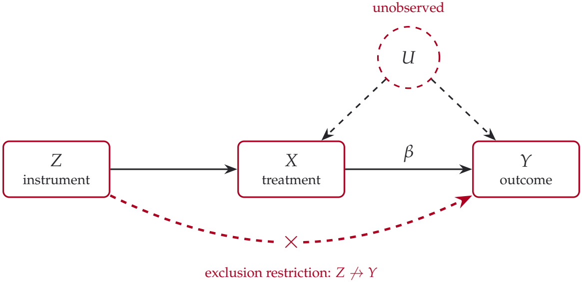
$\text{Cov}(z_i, x_i) \neq 0$ — the instrument actually shifts the treatment. Checked with the *first stage*.

Exogeneity / exclusion (not testable)

$\text{Cov}(z_i, \varepsilon_i) = 0$ — equivalently the **orthogonality (moment) condition** $\mathbb{E}[z_i \, \varepsilon_i] = 0$. The instrument affects y *only* through x, not through any other channel. Defended with economic argument, not a test.

The art of IV is almost entirely about **defending exogeneity**.

The logic in one picture



- **Relevance:** $Z \to X$ — the instrument moves the treatment.
- **Confounding:** U (unobserved) drives both X and Y — this is *why OLS is biased*.
- **Exclusion:** there is **no** arrow $Z \to Y$ (and none $U \to Z$) — so Z touches Y *only* through X. IV uses just the Z-driven part of X, which is clean of U.

The IV / Wald estimator

With a single instrument, take the covariance of the structural equation with z:

$$\mathrm{Cov}(z,y) = \beta\,\mathrm{Cov}(z,x) + \underbrace{\mathrm{Cov}(z,\varepsilon)}_{=0}.$$

Solving for β gives the **IV estimator**:

$$\boxed{\hat{\beta}_{IV} = \frac{\mathrm{Cov}(z,y)}{\mathrm{Cov}(z,x)}}$$

Intuition: scale the instrument's effect on the outcome by its effect on the treatment. If z is binary, this is the **Wald estimator**, a ratio of differences in means; the name is also used for the IV ratio computed on a single contrast, such as one pair of years.

The numerator and denominator are each a **simple regression on** z: the **reduced form** (y on z) and the **first stage** (x on z). We study them next.

First stage, reduced form, and the IV ratio

The IV estimate is built from **two ordinary regressions on the instrument**:

First stage — does z move the treatment?

Regress the treatment on the instrument: $x_i = \pi z_i + \nu_i$.
The slope $\hat{\pi}$ measures **relevance**.

Reduced form — does z move the outcome?

Regress the outcome on the instrument: $y_i = \rho z_i + \eta_i$.
Under exclusion, z reaches y only via x, so $\rho = \beta \pi$.

Hence the **IV estimator** is just the **ratio** of the two:

$$\hat{\beta}_{IV} = \frac{\hat{\rho}}{\hat{\pi}} = \frac{\text{reduced form}}{\text{first stage}}.$$

If the instrument barely moves x ($\hat{\pi} \approx 0$), we divide by ~zero — the estimate explodes. This is the **weak-instrument** problem.

Two-Stage Least Squares (2SLS)

With controls w_i and possibly several instruments, the same idea is implemented in **two stages**:

- **Stage 1 (first stage):** regress x_i on the instrument(s) and controls, and keep the **fitted values**
$$x_i = \pi z_i + w_i\gamma + \nu_i \quad \longrightarrow \quad \hat{x}_i.$$
$\hat{x}_i$ is the part of x driven *only* by the instrument — purged of U.
- **Stage 2 (second stage):** regress y_i on $\hat{x}_i$ (not x_i!) and controls:
$$y_i = \beta\,\hat{x}_i + w_i\delta + \xi_i.$$

With a single instrument this reproduces exactly $\hat{\beta}_{IV} = \hat{\rho}/\hat{\pi}$. In matrix form, with Z the $n \times L$ matrix of the L instruments (controls partialled out) and $P_Z = Z(Z'Z)^{-1}Z'$ the projection onto them,
$$\hat{\beta}_{2SLS} = (X'P_Z X)^{-1}X'P_Z y \qquad \text{(stage 1 forms } \hat{X} = P_Z X\text{)}.$$

In practice, estimate 2SLS with a dedicated IV routine (Stata `ivreghdfe`; Python `feols` / pyfixest).

What does IV estimate? LATE and compliers

Take z and x **binary** (z = an offer of the treatment). With **heterogeneous** effects β_i, IV recovers a *local* effect, not the average one, under one extra assumption:

Monotonicity (no defiers)

If the instrument affects treatment in one direction — positively (or negatively) — then **no** unit reacts in the opposite direction (no defiers).

- A **complier** is a unit whose treatment the instrument actually moves (treated only when encouraged, untreated otherwise).
- Under relevance, exclusion and monotonicity, the Wald estimator identifies the **Local Average Treatment Effect** — the average effect **for compliers**:

$$\hat{\beta}_{IV} \xrightarrow{p} \mathbb{E}[\beta_i \mid i \text{ complier}] = \text{LATE}.$$

- So IV gives the effect **for compliers**, not the population — different instruments give different LATEs.

*Imbens & Angrist (1994). This is why *negative* shift-share weights (later) warn of heterogeneity.*

Several instruments for one regressor

So far we used a single instrument z for the endogenous x. But we often have **several** valid instruments $z_1, \ldots, z_L$ for the *same* x — e.g. for the return to schooling, both distance to college and quarter of birth (defended next).

Excluded vs. included. The z_ℓ are the **excluded instruments**: they appear in the **first stage** (they help predict x) but are **excluded from the outcome equation**. The controls w_i, which appear in *both* equations, are the **included instruments**.

The **exclusion restriction** is exactly the claim that the excluded instruments have **no direct effect** on y — they reach y only through x.

The art of IV: defending each instrument

Example: returns to schooling

x = years of schooling, y = wages. Candidate instruments: z_1 = distance to the nearest college, z_2 = quarter of birth.

For each candidate, two questions: is it **relevant** (does it move schooling? testable in the first stage), and is it **excluded** (does it reach wages only through schooling? not testable, to be argued)?

z_1 **(distance to college, Card 1995).** *Relevant?* proximity cuts the cost of attending ⇒ more schooling (visible in the first stage). *Excluded?* it reaches wages only through schooling — credible **only after** controls for region, urbanicity and family background (college towns may differ economically).

z_2 **(quarter of birth, Angrist–Krueger 1991).** *Relevant?* school-entry cutoffs and a compulsory-schooling age let those born early leave with less schooling. *Excluded?* birth timing is as-good-as random — but the first stage is **tiny**, a **weak instrument** (Bound–Jaeger–Baker 1995).

The first stage with several instruments

All excluded instruments (together with the controls) enter the **first stage** at once:

First stage

$$x_i = \pi_1 z_{1i} + \cdots + \pi_L z_{Li} + w_i\gamma + v_i \quad \longrightarrow \quad \hat{x}_i.$$

- 2SLS forms the fitted value $\hat{x}_i$ — the **single best linear combination** of all the instruments — pooling their information automatically.
- The **second stage is unchanged**: regress y_i on $\hat{x}_i$ (and the controls).
- Relevance is now a **joint** question — do the instruments *together* move x?

That joint question is what the **first-stage** F (next slide) answers.

Are the instruments strong enough? The first-stage F

First-stage F test

Test the joint null that the excluded instruments do *not* explain the treatment: $H_0: \ \pi_1 = \cdots = \pi_L = 0$. With a *single* instrument ($L = 1$) it is just the squared first-stage t: $F = t^2$.

- **Weak instruments** ($=$ low F): 2SLS is **biased toward OLS** and its CIs **over-reject —** even with a valid instrument.
- **Rule of thumb:** $F > 10$ (Staiger & Stock, 1997).

Stock–Yogo (2005): the rule of thumb made rigorous

Critical values for the first-stage F to formally reject "weak instruments" at a stated tolerance (e.g. 2SLS bias $\leq$ 10% of OLS bias); the threshold *rises* with the number of instruments.

- Under heteroskedasticity or clustering, use the **Kleibergen–Paap** F, the first-stage F with robust standard errors. F tests **relevance**; Hansen's J (later) tests **validity**.

CIs that survive weak instruments: Anderson–Rubin

When the first-stage F is low, 2SLS t-based intervals are **unreliable**. The **Anderson–Rubin (AR)** test repairs inference on β.

The idea

To test $H_0 : \beta = \beta_0$, impose it and ask whether the instrument still predicts the imposed residual:

$$y_i - \beta_0 x_i = z_i'\theta + u_i, \qquad H_0 : \ \theta = 0.$$

If β_0 is true, $y_i - \beta_0 x_i$ is exogenous, so z has **no** power.

- The **AR set** = all β_0 not rejected; correct coverage **at any instrument strength**. With one instrument it is **exact**.
- It can be **wider**, unbounded, or empty — itself informative about (non-)identification.
- **Lee, McCrary, Moreira & Porter (2022):** $F > 10$ is too lenient for honest 5% inference — prefer **AR** (or their tF correction).

Section 2

GMM

What is a "moment"?

IV is a **method of moments**. A **moment** is the expectation of a function of the data: the mean $\mathbb{E}[x]$, a covariance $\mathbb{E}[xy]$.

The method of moments, in one line

A model implies that certain moments take known values. Replace the population expectation $\mathbb{E}[\cdot]$ by its **sample average** $\frac{1}{n}\sum_i$, and solve for the parameters.

IV is exactly this. Exogeneity is a population **moment condition**:

$$\mathbb{E}[z_i\, \varepsilon_i(\beta)] = 0, \qquad \varepsilon_i(\beta) = y_i - \beta x_i$$

("the instrument is uncorrelated with the error"; variables are demeaned, so the intercept is dropped). We estimate β by making the **sample** version hold:

$$\frac{1}{n}\sum_i z_i(y_i - \beta x_i) = 0 \implies \hat{\beta}_{IV} = \frac{\mathrm{Cov}(z,y)}{\mathrm{Cov}(z,x)}.$$

One equation, one unknown β — solved **exactly**.

One instrument, or several?

One instrument (just-identified): one moment, one unknown $\Rightarrow$ exact solution (the case above).

Two instruments for the *same* β. In our schooling example ($z_1 =$ distance to college, $z_2 =$ quarter of birth), each is its own moment condition:

$$\mathbb{E}[z_{1i}\,\varepsilon_i] = 0 \qquad \text{and} \qquad \mathbb{E}[z_{2i}\,\varepsilon_i] = 0.$$

- Each *alone* gives $\hat{\beta}_1 = \dfrac{\mathrm{Cov}(z_1, y)}{\mathrm{Cov}(z_1, x)}$, $\hat{\beta}_2 = \dfrac{\mathrm{Cov}(z_2, y)}{\mathrm{Cov}(z_2, x)}$.
- In finite samples $\hat{\beta}_1 \neq \hat{\beta}_2$: **two equations, one unknown — over-identification** ($L = 2 > 1$).

$\rightarrow$ **GMM** answers both: **(1)** how to *combine* them? **(2)** do they *agree* (a test of validity)?

The GMM estimator

Stack the L sample moments, with $z_i = (z_{1i}, \ldots, z_{Li})'$,

$$\bar{g}(\beta) = \frac{1}{n}\sum_i z_i(y_i - \beta x_i),$$

and choose β to make them **jointly as close to zero as possible**, in the metric of a weight matrix W:

$$\boxed{\hat{\beta}_{GMM} = \arg\min_{\beta}\ \bar{g}(\beta)'\, W\, \bar{g}(\beta)}$$

- W decides **how much each moment counts**. For these linear moments the minimiser has a closed form: $\hat{\beta}_{GMM} = (x'ZWZ'x)^{-1}x'ZWZ'y$, with Z the $n \times L$ instrument matrix.
- **2SLS is the case** $W = (Z'Z)^{-1}$**.** The weight matrix is what will reconnect the Bartik 2SLS to a GMM on the shares (shares view). *Efficient* (two-step) GMM sets $W = \hat{S}^{-1}$, the inverse of the moments' covariance.
- Over-identification also gives a **specification test** of instrument validity, Hansen's J (next slide).

Are the instruments valid? The over-identification test

Over-identified ($L >$ #endog), valid instruments should **agree** up to noise — disagreement flags an invalid one.

Sargan–Hansen J statistic

$$J = n\,\bar{g}(\hat{\beta})'\,\hat{S}^{-1}\,\bar{g}(\hat{\beta}) \xrightarrow{d} \chi^2_{L-\#\text{endog}}, \qquad H_0: \mathbb{E}[z_i \varepsilon_i] = 0.$$

Sargan (homoskedastic) / Hansen (robust); dof $= L - 1$ with one endogenous x.

- **Reject** $\Rightarrow$ an over-identifying restriction fails: an instrument is invalid (or the model is misspecified).
- **Just-identified** ($L =$ #endog) $\Rightarrow$ 0 dof: exclusion **cannot be tested**.
- A **joint**, low-power test: it won't say *which* fails, and a pass is **reassuring, not proof**.

Dynamic panels: GMM with lagged instruments

The most common GMM in applied work is **dynamic panel GMM**, used when a **lagged outcome** sits on the right-hand side:

$$y_{it} = \rho\, y_{i,t-1} + \beta x_{it} + f_i + u_{it}.$$

- **Within breaks here** (Nickell 1981): sweeping out f_i leaves $y_{i,t-1}$ correlated with the demeaned error — biased, badly for small T.
- **Arellano–Bond (1991):** first-difference to remove f_i, then instrument $\Delta y_{i,t-1}$ with **deeper lagged levels** $y_{i,t-2}, y_{i,t-3}, \ldots$
- **Blundell–Bond (1998)** "system GMM": also use lagged *differences* as instruments for the levels equation (needs an extra stationarity assumption).

…and why dynamic GMM is questionable

Not a default choice

In the **dynamic** case especially, GMM becomes a **black box**: fragile results, and diagnostics that can mislead.

- **Instruments proliferate** with $T \Rightarrow$ it overfits $y_{i,t-1}$, drags $\hat{\beta}$ toward the biased within estimate, and inflates the Hansen J toward $p \approx 1$ — a meaningless "pass" (Roodman 2009).
- **Researcher choices** (lag depth, "collapsing" the instrument set to fewer lags, one- vs two-step, the Windmeijer correction of two-step standard errors) make a result easy to *fish*.
- **Weak** when y is persistent: lagged levels barely move the differences.

Shift-share is different: *not* a lag-instrument GMM, but an instrument from **observed** shares and shocks with an **economic story** for exogeneity. Goldsmith-Pinkham et al. (2020) even show it *is* GMM — but a **transparent** one whose identifying variation we can see and defend.

Section 3

Shift-share IV

Where do good instruments come from?

The binding constraint is almost never *relevance* — it is *exclusion*.

- A valid instrument needs a credible argument for why it affects y **only** through x.
- Natural experiments (lotteries, policy discontinuities, weather…) are scarce.

Where do good instruments come from?

The binding constraint is almost never *relevance* — it is *exclusion*.

- A valid instrument needs a credible argument for why it affects y **only** through x.
- Natural experiments (lotteries, policy discontinuities, weather…) are scarce.

A recurring idea in applied work: many outcomes at the level of a *region*, *city*, *or firm* respond to **national or global shocks**, but each unit is exposed to those shocks **differently**, according to its **pre-existing composition**.

That structure — common shocks $\times$ heterogeneous exposure — is exactly the **shift-share** instrument. It is the workhorse of labor, trade, macro, public, and urban economics.

Motivating setting: local labor markets

Classic question (Bartik 1991): what is the elasticity of local **wages** y_i to local **labor demand** x_i across regions i?

- OLS is biased: a local labor-supply shock (e.g. immigration) moves both employment and wages — $\mathrm{Cov}(x_i, \varepsilon_i) \neq 0$.
- We need an instrument that shifts **local labor demand** for reasons unrelated to local supply.

Idea: a region's industries grow or shrink with *national* industry trends; a region tied to booming industries sees demand rise — regardless of its local supply conditions.

Notation

Throughout the shift-share part:

symbol	meaning
i	unit of analysis (region, city, firm)
$k = 1, \ldots, K$	sectors / origins / "shocks"
s_{ik}	**share**: exposure of unit i to k (e.g. industry k's employment share), predetermined
g_k	**shift**: the common shock to k (e.g. national growth of industry k)
$x_i,\ y_i$	endogenous treatment and outcome

Shares are typically (not always) non-negative and **sum to one**: $\sum_k s_{ik} = 1$.

Construction of the instrument

The **shift-share** instrument (traditionally the **Bartik** instrument) interacts predetermined shares with common shifts:

$$\boxed{z_i \;=\; \sum_{k=1}^{K} s_{ik}\, g_k}$$

- *Shares* s_{ik} — measured at a **baseline** period (before the shocks).
- *Shifts* g_k — common to all units (national / global).

z_i is the **predicted** growth of unit i if each of its sectors grew at the national rate. It is then used as an instrument for the endogenous x_i.

A worked mini-example

Two sectors (manufacturing, services), two regions. Baseline shares:

region	manuf.	services
A	0.70	0.30
B	0.20	0.80

National shifts: $g_{\text{manuf}} = -10\%$ (China shock), $g_{\text{serv}} = +5\%$.

$$z_A = 0.7(-10) + 0.3(5) = -5.5\%, \qquad z_B = 0.2(-10) + 0.8(5) = +3.0\%.$$

The instrument predicts region **A contracts**, region **B expands** — purely from *national* trends × *local* exposure, not from local supply shocks.

Three canonical applications

study	unit i	shares s_{ik}	shifts g_k
Bartik (1991)	region	industry employment shares	national industry growth
Card (2001)	city	origin-k migrant share, 1980	national inflow from origin k
Autor et al. (2013)	region	industry employment shares	Chinese imports (other high-income countries)

Same skeleton everywhere: **predetermined exposure** × **common shock**. The designs differ in *which component* carries the identifying assumption — the heart of this lecture.

Leave-one-out shifts

A subtle but crucial point: if the shift g_k is built from a national aggregate that **includes unit i itself**, then z_i mechanically contains i's own outcome shock $\Rightarrow$ correlation with ε_i.

Fix: construct **leave-one-out** shifts that exclude unit i (or, as in Autor et al., measure the shock in *other countries* entirely):

$$g_k^{(-i)} = \text{growth of sector } k \text{ excluding unit } i.$$

This breaks the mechanical link and is standard practice.

The central question

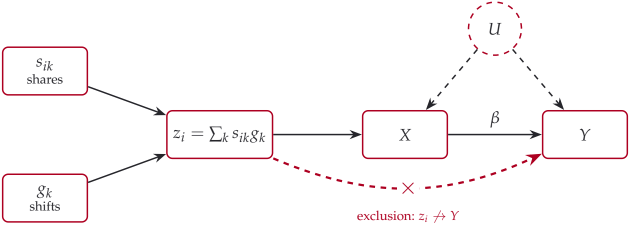
Why is z_i a valid instrument?

$z_i = \sum_k s_{ik} g_k$ has **two** ingredients. Exogeneity can be argued to come from *either* one:

- **From the shares s_{ik} —** units differ in *exposure*, and that exposure is as-good-as-random (conditional on controls). $\rightarrow$ **Goldsmith-Pinkham et al. (2020)**.
- **From the shifts g_k —** the *shocks* are as-good-as-random, even if exposure is endogenous. $\rightarrow$ **Borusyak et al. (2022)**.

These are **different research designs** with different assumptions, diagnostics, and standard errors. We take each in turn.

The two routes, in one picture



Both ingredients feed z_i; its exogeneity (clean of U) can be earned from **either**:

- **Goldsmith-Pinkham et al. (shares):** the shares are as-good-as-random $\Rightarrow \mathbb{E}[s_{ik}\varepsilon_i] = 0$.
- **Borusyak et al. (shifts):** the shifts are as-good-as-random $\Rightarrow$ valid *even if shares are endogenous*.

Section 4

Shares view

The big idea

Goldsmith-Pinkham et al. (2020) ask: *what is a Bartik regression actually doing?*

Numerical equivalence (their Proposition 1.1)

The 2SLS estimator using the single instrument $z_i = \sum_k s_{ik} g_k$ is *numerically identical* to a GMM estimator that uses the K industry shares $\{s_{ik}\}_k$ as **separate instruments**, with a weight matrix built from the shifts.

Formally, with z the instrument, x the treatment, y the outcome (stacked over units), S the matrix of shares and W a weight matrix:

$$\hat{\beta}_{Bartik} = \frac{z'y}{z'x}, \qquad \hat{\beta}_{GMM} = \frac{x'SWS'y}{x'SWS'x}.$$

With the rank-one weight matrix $W = gg'$ (the outer product of the K-vector of shifts g) and $z = Sg$ (the $n \times K$ shares matrix S times g), the two coincide: $S\,gg'\,S' = (Sg)(Sg)' = zz'$, so $\hat{\beta}_{GMM} = \hat{\beta}_{Bartik}$.

Why this reframes identification

If Bartik = "shares as instruments", then the **exclusion restriction lives on the shares**:

$$\mathbb{E}[s_{ik}\,\varepsilon_i] = 0 \quad \text{for every } k.$$

The shifts g_k are *not* instruments here — they only **combine** the K share-instruments into one. The identifying content is: *is each exposure share uncorrelated with the unobservable?*

This is the natural reading when you have **few, dominant sectors** and tell a story like "places more exposed to industry k form a clean comparison group."

Opening the black box: Rotemberg weights

Building on Rotemberg (1983), Goldsmith-Pinkham et al. decompose the over-identified estimator into the K **just-identified** estimators, one per share-instrument:

Proposition 3.1 (Goldsmith-Pinkham et al.)

$$\hat{\beta}_{Bartik} = \sum_{k=1}^{K} \hat{\alpha}_k \, \hat{\beta}_k, \qquad \sum_k \hat{\alpha}_k = 1,$$

where the just-identified estimate and the **Rotemberg weight** are

$$\hat{\beta}_k = \left(s_k' x\right)^{-1} s_k' y, \qquad \hat{\alpha}_k = \frac{g_k \, s_k' x}{\sum_{k'} g_{k'} \, s_{k'}' x}.$$

(Here x, y are residualized on the controls; $s_k = \{s_{ik}\}_i$, the k-th share column.)

So Bartik is a **weighted average of single-share IV estimates**.

Reading the Rotemberg weights

The weight $\hat{\alpha}_k$ tells you **how much industry k drives the headline estimate** — and how sensitive $\hat{\beta}$ is to a violation of exogeneity in that one share. Practical lessons from Goldsmith-Pinkham et al.:

- Weights measure **sensitivity to misspecification** (Andrews–Gentzkow–Shapiro 2017): bias in a high-$\hat{\alpha}_k$ share passes strongly into $\hat{\beta}$.
- The weights **sum to one but can be negative** — an *affine*, not convex, average; a warning for the LATE interpretation under heterogeneous effects.
- The **shifts g_k explain little** of the weights ($< 1\%$ in their example) — "the variation comes from the shocks" is often *wrong*.
- In their application, **5 industries hold** $> 40\%$ of the weight (oil & gas on top).

Lineage: that IV recovers a **weighted average of heterogeneous effects** dates to Imbens & Angrist (1994) / Angrist & Imbens (1995); negative weights under many instruments are sharpened by Mogstad, Torgovitsky & Walters (2021).

The identifying assumption (shares view)

Share exogeneity

The exposure shares s_{ik} are uncorrelated with the unit-level unobservable ε_i (conditional on controls).

When outcomes are in changes, this is a **parallel-trends** condition: absent the shock, units with high and low exposure to k would have moved alike (the identifying assumption of the difference-in-differences design, next chapter). A **pre-trend** test checks it on the years before the shock. The shift-share pools **many such comparisons**, one per share; inference relies on **many units** (large N).

Diagnostics, for the high-weight shares:

- correlate them with observable confounders and baseline characteristics;
- test for **pre-trends** in their comparisons and in z_i overall;
- report **over-identification** tests and the spread of the $\hat{\beta}_k$ (`bartik_weight` package).

Section 5

Shifts view

The big idea

Borusyak et al. (2022) start from a different intuition:

A share-weighted average of random shifts is itself as-good-as-random

Even if the shares s_{ik} are *endogenous* (exposed units differ systematically), as long as the shocks g_k are as-good-as-randomly assigned across many sectors, the instrument $z_i = \sum_k s_{ik} g_k$ is mean-independent of ε_i on average.

Example: skill-intensive regions may attract particular immigrants (endogenous shares), but if subsidies are randomized across many high- and low-skill industries, exposed regions get **typical** instrument values.

The shift-level "translation"

Borusyak et al. show the shift-share IV is a **translation device**: the location-level IV estimator is numerically equal to a **shock-level IV regression** that uses g_k *directly* as the instrument.

- Aggregate the outcome and treatment to the **shock level** with exposure weights: $\bar{y}_k = \sum_i s_{ik} y_i \big/ \sum_i s_{ik}$, likewise $\bar{x}_k$; the importance of shock k is $\omega_k = \sum_i s_{ik} \big/ \sum_{k'} \sum_i s_{ik'}$, so that $\sum_k \omega_k = 1$.
- Run the IV of $\bar{y}_k$ on $\bar{x}_k$, weighted by ω_k, instrumenting with g_k.

The **same number** comes out — but now the assumption is stated on the **shocks**, and there are as many "observations" as there are **shocks** K, not units.

How many shifts do you really have?

Because each shift carries an importance weight ω_k, the **effective sample size** is governed by the concentration of those weights:

Effective number of shifts

$$K_{\text{eff}} = \frac{1}{\sum_k \omega_k^2}, \qquad \sum_k \omega_k^2 = \text{the Herfindahl index (HHI) of the exposure weights.}$$

- If a few sectors dominate exposure, K_{eff} is **small** — the design rests on a handful of shocks and is fragile.
- The shifts approach needs K_{eff} **large** so a law of large numbers kicks in.

The identifying assumptions (shifts view)

Two requirements

(1) Quasi-random shocks (shock orthogonality): the g_k are as-good-as randomly assigned and affect y only through x — mean-independent of the exposure-weighted unobservable.

(2) Many, dispersed shocks: K_{eff} large (low exposure Herfindahl), shocks mutually uncorrelated.

Notably, **no assumption on the shares** is required — they may be endogenous. This fits "generic" industry shares (as in Autor et al. 2013) where share exogeneity would be implausible.

Controls and diagnostics (shifts view)

Make shocks conditionally as-good-as-random with the right controls:

- If shares do **not** sum to one, **control for the sum of shares** $\sum_k s_{ik}$.
- For a shock-level confounder q_k (e.g. skill intensity), control for its **shift-share aggregate** $\sum_k s_{ik} q_k$ — same shares, applied to q_k.

Diagnostics:

- **Balance tests** at the shock level: regress g_k on shock-level characteristics.
- Report K_{eff} and exposure-weighted summary statistics.
- Use leave-one-out / externally-measured shifts to kill mechanical correlation.
- The `ssaggregate` package builds the shock-level dataset.

Standard errors also need special care under shared exposure — the next section.

Section 6

Inference

Why standard errors are wrong

A subtle problem (Adão et al. 2019): **two regions with similar exposure share vectors get similar instrument values** z_i, because z_i is the *same* linear combination of common shocks.

⇒ regression residuals are **correlated across units that look unrelated** (even in different states/clusters).

Conventional, heteroskedasticity-robust, and even region-clustered standard errors **ignore this** and badly **over-reject —** a nominal 5% test can reject 30–50% of the time.

The fix: exposure-robust standard errors

Practical guidance

Use standard errors that are robust to cross-unit correlation *induced by shared exposure*:

- **Exposure-robust** estimators of Adão et al. (2019) — valid under the exposure structure;
- equivalently, **shock-level (Borusyak et al.) standard errors** from the shift-level IV regression cluster naturally **at the level of the shocks** k;
- both are implemented in the authors' Stata/R packages.

Rule of thumb: the unit of "independent variation" is the **shock**, not the region — cluster your thinking (and SEs) accordingly.

Section 7

Shares or shifts?

Which assumption fits your design?

Ask yourself:

1. **Can I defend a few shares as exogenous?** Are they *tailored* (they mediate only the shock of interest, like origin-specific migrant shares)? → **shares view**.
2. **Are my shares "generic"** (industry composition that proxies exposure to *many* unobserved shocks)? Then share exogeneity is implausible → lean on **exogenous shifts**.
3. **Do I have many, plausibly-random shocks** with low exposure HHI? → **shifts view** is available and powerful.

In practice: report **both** sets of diagnostics, and always use **exposure-robust standard errors**.

Two paths, one instrument

	Goldsmith-Pinkham et al.	**Borusyak et al.**
Exogenous	the **shares** s_{ik}	the **shifts** g_k
Shares may be endogenous?	no	**yes**
Asymptotics in	number of units	number of **shocks** K
Best when	few, dominant sectors; "tailored" shares	many small shocks; "generic" shares
Key diagnostic	Rotemberg weights, pre-trends	balance of g_k, K_{eff}
Inference	over-id tests; exposure-robust SE	shock-level / Adão et al. SE
Canonical	Card (2001) migration enclaves	Autor et al. (2013) China shock

Section 8

Applied

Setting and data

The applied session estimates how much an input bought by farms responds to the farm's revenue: y is the log of input spending, x the log of crop revenue, endogenous.

- **Units and indices.** i = farm, c = crop (twelve crops), t = year. The share s_{ic} is crop c's part of the farm's crop revenue in its first two observed years; the shift is the national price of crop c in year t, $\log P_{ct}$.
- **The instrument is a panel,** $z_{it} = \sum_c s_{ic} \log P_{ct}$, used with farm and year fixed effects: within a farm it moves over time only because prices move.
- **Objects in the code.** `df`: one row per farm and year, columns `id`, `annee`, `x`, `y`, `prod_veg` (total crop output) and one revenue column per crop. `prix`: one row per crop and year, columns `culture`, `annee`, `log_prix`. `crops`: the list of the twelve crop columns.

```
import pandas as pd, pyfixest as pf
```

Build the two ingredients: shifts and shares

Four lines turn a price file and the panel into the shifts $\log P_{ct}$ and the shares s_{ic}.

```
lp = prix.pivot(index="annee", columns="culture",
                values="log_prix")                              # 1
base = df.groupby("id").head(2)                                 # 2
tot = base.groupby("id")[crops + ["prod_veg"]].sum()            # 3
shares = tot[crops].div(tot["prod_veg"], axis=0)                # 4
```

- **1** — the shifts as a table with one row per year and one column per crop: `pivot` reshapes the long price file into that grid. `lp.loc[2007, "colza"]` is a price.
- **2** — the **first two rows** of each farm, its two earliest years: the shares must be measured before the shocks they will be combined with.
- **3** — sums, over those two years, the revenue of each crop and total output.
- **4** — divides each crop's revenue by total output, row by row (`axis=0`): the share of each crop in the farm's output, one row per farm, one column per crop.

Combine them into the instrument z_{it}

Five lines attach the shares to every year of the farm and sum shares times prices.

```
pan = df.merge(shares, left_on="id", right_index=True)      # 1
pan["z"] = 0.0                                              # 2
for c in crops:                                             # 3
    pan[f"z_{c}"] = pan[c] * lp.loc[pan["annee"], c].values # 4
    pan["z"] += pan[f"z_{c}"]                               # 5
```

- **1** — joins the shares to the panel: `df` has a column `id`, `shares` has the farm as its row index, so each farm's twelve shares are copied onto all of its rows.
- **2** — starts the instrument at zero.
- **3, 4** — for each crop, the contribution share × log price is stored in a column z_<crop>, used later for the Rotemberg weights. `pan[c]` is the farm's share of crop `c`; `lp.loc[pan["annee"], c]` picks, for each row, the price of `c` in that row's year; `.values` drops the index so that the two line up row by row.
- **5** — adds it to `z`. After the loop, `z` is $\sum_c s_{ic} \log P_{ct}$.

First stage, reduced form, 2SLS

Four lines estimate the three regressions of the chapter and the first-stage F.

```
cl = {"CRV1": "id"}
fs = pf.feols("x ~ z | id + annee", pan, vcov=cl)                # 1
rf = pf.feols("y ~ z | id + annee", pan, vcov=cl)                # 2
iv = pf.feols("y ~ 1 | id + annee | x ~ z", pan, vcov=cl)        # 3
F  = (fs.coef()["z"] / fs.se()["z"]) ** 2                        # 4
```

- **1** — the first stage: the endogenous regressor on the instrument, farm and year fixed effects absorbed. Its coefficient is $\hat{\pi}$.
- **2** — the reduced form: the outcome on the instrument. Its coefficient is $\hat{\rho}$.
- **3** — 2SLS in one call. The formula has **three parts** separated by `|`: the outcome and its exogenous regressors (`1` means none besides the fixed effects), the fixed effects, then `endogenous ~ instrument`. The coefficient on `x` is $\hat{\rho}/\hat{\pi}$.
- **4** — with one instrument, the first-stage F is the square of its t.
- `cl` — standard errors clustered by farm; the exposure-robust errors of the inference section are not available in pyfixest.
- To test an elasticity of 1 rather than 0: `(iv.coef()["x"] - 1) / iv.se()["x"]`.

One shock at a time: the Wald estimator

Four lines compute the ratio of two differences on a single pair of years.

```
w = pan[pan["annee"].isin([2006, 2007])]                          # 1
w = w.pivot(index="id", columns="annee",
            values=["y", "x", "z"]).dropna()                      # 2
d = pd.DataFrame({"dy": w["y"][2007] - w["y"][2006],
                  "dx": w["x"][2007] - w["x"][2006],
                  "dz": w["z"][2007] - w["z"][2006]})             # 3
wald = pf.feols("dy ~ dz", d).coef()["dz"] \
     / pf.feols("dx ~ dz", d).coef()["dz"]                        # 4
```

- **1** — the two years around the shock.
- **2** — one row per farm, one column per year and variable; a farm missing a year gets a gap, dropped with `.dropna()`.
- **3** — the change of each variable between the two years, for each farm.
- **4** — reduced form over first stage on those changes: the Wald estimator, identical to 2SLS with one instrument. Its first-stage F tells whether the shock spread farms enough.

Which crops carry the estimate, and a placebo

Five lines compute the Rotemberg weights and build the placebo, next year's instrument.

```
def r(v): return pf.feols(f"{v} ~ 1|id + annee", pan).resid() # 1
xt = r("x")                                                    # 2
num = {c: r(f"z_{c}") @ xt for c in crops}                     # 3
alpha = {c: num[c] / sum(num.values()) for c in crops}         # 4
pan["z_lead"] = pan.groupby("id")["z"].shift(-1)               # 5
```

- **1** — a function that regresses a variable on the fixed effects only and returns the residuals: the variable **net of** farm and year effects. **2** applies it to `x`.
- **3** — for each crop, the covariance between its contribution to the instrument, `z_c` $= s_{ic} \log P_{ct}$, and `x`; `@` is the dot product of two vectors.
- **4** — each covariance over their sum: the Rotemberg weight α_c. They sum to one; a negative one is possible.
- **5** — `z` on the **next row of the same farm**, `shift(-1)` moving values up one row inside each farm: next year's prices. Added to the reduced form, a significant coefficient would mean farms react to prices before they occur.

References I

Adão, Rodrigo, Michal Kolesár, and Eduardo Morales. 2019. "Shift-Share Designs: Theory and Inference." *The Quarterly Journal of Economics* 134 (4): 1949–2010. https://doi.org/10.1093/qje/qjz025.

Borusyak, Kirill, Peter Hull, and Xavier Jaravel. 2022. "Quasi-Experimental Shift-Share Research Designs." *The Review of Economic Studies* 89 (1): 181–213. https://doi.org/10.1093/restud/rdab030.

Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift. 2020. "Bartik Instruments: What, When, Why, and How." *American Economic Review* 110 (8): 2586–624. https://doi.org/10.1257/aer.20181047.